An Intuitive Computer Forensic Method by Timestamp Changing Patterns

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***Abstract*—**This proposes an intuitive computer forensic method by timestamp changing patterns of operations on file in Windows NTFS file system. It categorized by seven file operations and has ten distinguishable patterns by their timestamp changes. The distinct timestamp changing patterns make decision on identifying what kind of file operation is performed. Some patterns are easily identified by their distinct timestamp feature intuitively, and some patterns are needed past timestamp to identify the file operation clearly, and some patterns have ambiguity with similar timestamp patterns. With some performed cases, the forensic method is tested and presented for its usage.

Digital forensics, timestamp changing pattern, intuitive forensic , event reconstruction, NTFS filesystem

1. Introduction

Timestamp evidence is a fundamental ingredient of many forensic computing examinations to reconstruct the event, and the determination of event times is an important and difficult task in computer forensics. MAC times are pieces of file system metadata which record when certain events pertaining to a computer file occurred most recently[1].

There are many researches on timelines based on file system time. Casey[2] has indicated that MAC times analysis is necessary to the reconstruction of digital events. MAC timestamps record a file's most recent modification, access, and creation times. By reconstructing these on a timeline, forensic investigators can find filesystem activity, and computer usage of a particular time. An investigator can also draw a historical plot of filesystem activity per time period.[2,3]. Boyd and Forster discussed time structure and their use in Microsoft Internet Explorer with local and UTC time translation issues[4]. Chow et. al. presented behavioral characteristics of MAC times on NTFS file system so that validation basis for temporal analysis in event reconstruction models can be formulated[5]. Stevens has introduced a clock model that can account for the various factors that affect the behavior of digital clocks such as those used in computers and other digital electronic devices[6]. Willassen presents a hypothesis-based investigation methods to solve the problems of timestamps manipulation and a clock that is erroneous or improperly adjusted[7].

J. Olsson and M. Boldt[8] developed a computer forensic timeline visualization tool that the way of visualizing the evidence allows the investigators to find coherent evidence faster and more intuitively. C. Hargreaves and J. Patterson[9] proposes a technique that can automatically reconstruct high-level events from set of low-level events, analysing patterns automatically. They described a framework that extracts low-level events to a SQLite backing store, the provenance of any high-level events is also preserved, and the raw data that caused the low-level event to be initially created can also be viewed.

Recently, Cho[10] proposed a method for detecting timestamp forgery in NTFS filesystem. Log records operate on files leave large amounts of information in the $LogFile that can be used to reconstruct operations on the files and also used as forensic evidence. If the past timestamps can be found before any changes to the file are made, this could act as evidence of a file time forgery. Rule sets for detecting timestamp forgery based on using difference comparison between changes in timestamp patterns by the file time change tool and normal file operations is provided, and the forensic rule sets for ".txt", ".docx" and ".pdf" file types is applied for forensic cases.

There are some forensic researches on other file system. Kevin D. Fairbanks[11] presented a research that a low-level study and analysis of Ext4 file system data structures, which includes descriptions of extents, extent trees, directory indexing HTrees, and flex block groups. The paper provided a more comprehensive analysis with

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the forensics community in mind and states some brief implications of the file system behavior with respect to data recovery. Ming Xu[12] et. al. showed that the experimental results under Linux operating system show that the proposed method can correctly reconstruct file system, recover file and file traces from YAFFS2; and experiments conducted on physical images of Android phones show that our method can be applied to real scenarios. A. Marrington, et. al.[13] developed a tool, implementing techniques for detecting contradictory and missing events in the history of the computer system. The experiments with this software demonstrate that the techniques proposed can be used successfully to detect temporal inconsistencies in a computer activity timeline. The automatic detection of inconsistencies which might indicate deliberate tampering could assist a human investigator in a subsequent manual examination of the system.

In this paper, an intuitive computer forensic method by timestamp changing patterns of operations on file in Windows NTFS file system is proposed. In Chap. 3, ten timestamps pattern that is categorized by seven file operations and has ten distinctive patterns by their timestamp changes will be explained, and conduct analyses of ten timestamp patterns for performing an intuitive timestamp forensic method only with current timestamps. With some performed cases, the forensic method is tested and presented for its effectiveness in Chap. 4, and concluded in Chap 5.

2. Preliminaries

2.1. *Meanings of Symbol*

t_{op} : file operation performed time – A time when a user executes a file operation on an object file. Written without a superscript, it means a representative expression of each timestamp field.

t^C_{op}, t^W_{op}, t^E_{op}, t^A_{op} : file operation performed time with timestamp field name - Superscripts of four timestamps stand for C: "Creation", W: "Write", E: "MFT Entry modification" and A: "Access", respectively.

t_{src} : timestamps from a source file – A representative expression of he original timestamp from a source file before an operation is performed.

t_{SIsrc} : timestamps from $SI attribute of a source file - The timestamps come from $SI attribute of a source file to $FN attribute of a destination file.

U : unchanged timestamp – The timestamp have the same time value before and after a file operation performed. U is a representative expression of U^C, U^W, U^E, and U^A.

$SI/$FN : $STANDARD_INFORMATION attribute and $FILE_NAME attribute – These are abbreviated expressions of long attribute name of metafile in $MFT metafile of NTFS filesystem.

Δ : Delta time – Δ(delta) means an additional execution time to a file operation execution completed. Usually, this time is very small time compared to other file operation time. So, the Δ(delta) time can be considered $(t^E_{op}+\Delta)= t^E_{op}$.

3. An Intuitive Computer Forensic Method by Timestamp Changing Patterns

3.1. *Changes in Timestamps by File Operations*

We categorize ten patterns of file operations as follows: "Creation", "Copy", "Update-general", "Update-MS Word/Office", "Move-the same volume", "Move-the other volume", "Overwrite-source file left", "Overwrite-source file deleted", "File name change" and "File attribute change". These timestamps change patterns are based on the author's published research in reference [10] then we analyze the timestamp pattern of ten file operations for forensic purposes.

3.1.1 Creation

All of the timestamps of $SI and $FN "Creation" operation is set to the file creation operation time.

$$\$SI(t^C_{op}, t^W_{op}, t^E_{op}, t^A_{op})$$
$$\$FN(t^C_{op}, t^W_{op}, t^E_{op}, t^A_{op}) \quad (1)$$

The relationship among the timestamps in $SI and $FN fields is as follows:

$$(t^C_{op} = t^W_{op} = t^E_{op} = t^A_{op})_{\$SI}$$
$$= (t^C_{op} = t^W_{op} = t^E_{op} = t^A_{op})_{\$FN} \quad (2)$$

Timestamp Analysis : This operation has simple and clear timestamp pattern. And the time relationship in (2) is clear and easy to understand, and to decide the timestamps pattern is the result of "Creation" operation with only current timestamps. Because all of timestamp field in $SI and $FN attribute filled with operation time t_{op}, it is easy to decide what type of operation executed with

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only the current timestamps.

3.1.2 Copy

When we make a copied file by copy operation, all of the timestamps of $SI and $FN is set to the time t_{op} except write time of $SI. The t^W_{src} came from the "Write time" of a source file, and this field is distinguished feature of copy operation so this field is used to check whether the file is copied from some file or not.

$$\$SI(t^C_{op}, t^W_{src}, (t^E_{op}+\Delta), t^A_{op})$$
$$\$FN(t^C_{op}, t^W_{op}, t^E_{op}, t^A_{op}) \quad (3)$$

The time relationship in $SI and $FN fields is given as follows:

$$t^W_{src} << t_{op} \leq (t^E_{op}+\Delta),$$
$$\text{where if } \Delta \cong 0 \text{ then } t^E_{op} = t_{op} \quad (4)$$

Timestamp analysis : We need to look at timestamp pattern for the "Copy" operation. The second field of the timestamp in $SI, i.e. t^W_{src}, is a significant feature. The t^W_{src} originates from "Write tme" field in $SI attribute of a source file. As in (4), t^W_{src} precede the rest of timestamps, and all of timestamp field has the same t_{op} except t^W_{src}. This timestamp pattern is distinctive and unique, it is clearly to decide the "Copy" operation executed with only the current timestamps.

3.1.3 Content Update of General File

When an "Content Update of General File" operation is performed, "MFT entry modified time" field is changed to update operation time t_{op} and "Write time" is changed to $(t^W_{op}+\Delta)$ in $SI. But "Creation", "Access" field in $SI and all of $FN timestamps remain unchanged.

$$\$SI(U^C, (t^W_{op}+\Delta), t^E_{op}, U^A)$$
$$\$FN(U^C, U^W, U^E, U^A) \quad (5)$$

After the update operation is executed, the time relationship is as follows:

$$U << t_{op} \leq (t^W_{op}+\Delta),$$
$$\text{where if } \Delta \cong 0 \text{ then } t^W_{op} = t_{op} \quad (6)$$

Timestamp Analysis : The two timestamp, i.e. "Write time" and "MFT entry modified time", are chaged to $(t^W_{op}+\Delta)$ and t^E_{op} respectively. This timestamp pattern is distinctive and unique, mainly because of the two changed timestamp filelds. We can readily know U precedes t_{op} in (6), if these two timestamps relationship satisfy their time order then it is clearly to decide the "Content Update of General File" operation executed with only the current timestamps.

3.1.4 Content Update of MS Word/Office File

When an "Update of MS Word/Office File" is performed, the changes of timestamps are different to general files. Except "Creation" field in both $SI and $FN, all of timestamps in $SI and $FN are set to t_{op}.

$$\$SI(U^C, t^W_{op}, t^E_{op}, t^A_{op})$$
$$\$FN(U^C, t^W_{op}, t^E_{op}, t^A_{op}) \quad (7)$$

After the update operation executed, the time relationship in $SI and $FN fields will be as follows:

$$U << (t_{op})_{SI} = (t_{op})_{FN} \quad (8)$$

Timestamp Analysis : The three timestamp, "Write time", "MFT entry modified time" and "Access time", are chaged to t^W_{op}, t^E_{op} and t^A_{op} in $SI attribute, respectively, and same to in $FN. This timestamp pattern is distinctive and unique, mainly because of the six changed timestamp filelds. It is clearly to decide the "Update of MS Word/Office File" operation executed with only the current timestamps. We can readily know U precedes t_{op} in (8), if the relationship satisfy the time order, then it is clearly to decide the "Update of MS Word/Office File" operation executed.

3.1.5 Move within a Volume

When "Move within a Volume" operation is performed, "MFT entry modified time" is changed to t_{op} , and other else unchanged in $SI. And four timestamps in $FN are changed to t_{SIsrc}, respectively. (t_{SIsrc}) came from $SI of the original file.

$$\$SI(U^C, U^W, t^E_{op}, U^A)$$
$$\$FN(t^C_{SIsrc}, t^W_{SIsrc}, t^E_{SIsrc}, t^A_{SIsrc}) \quad (9)$$

The time relationship in $SI and $FN attributes is as follows:

$$U << t^E_{op}$$
$$t_{SIsrc} << t^E_{op} \quad (10)$$

where the relationship between U and t_{src} is not defined clearly, because these are dependent upon their initial states. This operation makes similar timestamp pattern to "Filename Change" operation in (17). Unfortunately, this pattern is not unique.

Timestamp Analysis : This operation changes a file name when the operation is completed. In this case, there

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is no change of the file lcoation, so a source and a destination of file are identical. In $FN attribute of (17), t^C_{SIsrc},t^W_{SIsrc},t^E_{SIsrc} and t^A_{SIsrc} in $FN attribute copied from its $SI attribute before the operation is performed. As in (18), ($U = t_{SIsrc}$) is a distinctive pattern, but it has similar pattern of "Move with a volume" operation in (9), unfortunately, this pattern is not unique.

3.1.6 Move to an Outside Volume

When "Move to an Outside Volume" operation is performed, "Creation" and "Write" field is changed to t^C_{src} and t^W_{src} respectively in $SI. And "MFT entry modified time" is changed to ($t^E_{op}+\Delta$), and "Access" field is changed to t^A_{op}. All of $FN fields is changed to the "Move to an Outside Volume" operation time t_{op}.

$$\$SI(t^C_{src}, t^W_{src}, (t^E_{op}+\Delta), t^A_{op})$$
$$\$FN(t^C_{op}, t^W_{op}, t^E_{op}, t^A_{op}) \quad (11)$$

The time relationship in $SI and $FN fields is as follows:

$$t_{src} << t_{op} \leq (t^E_{op}+\Delta),$$
$$\text{where if } \Delta \cong 0 \text{ then } t^E_{op} = t_{op} \quad (12)$$

Timestamp Analysis : This operation has simple and clear timestamp pattern comparatively. And the time relationship in (12) is clear and easy to understand, and to decide the timestamps pattern is the result of "Overwrite with source file left" operation with only current timestamps. Because six timestamp, include "MFT entry modified time" and "Access" field in $SI and All of four timestamp fields in $FN attribute. If we can secure the two timestmps also, then it will be perfect, but the current timestamps only are enough to analyze the pattern from what type of operation.

3.1.7 Overwrite with Source File Left

When "Overwrite with Source File Left" operation is performed, "MFT entry modified time" in $SI is changed to t^E_{op} and "Write time" in $SI is changed to t^W_{src} which comes from "Write time" of a source file. The remaining two fields are unchanged. $FN fields are all unchanged.

$$\$SI(U^C, t^W_{src}, t^E_{op}, U^A)$$
$$\$FN(U^C, U^W, U^E, U^A) \quad (13)$$

The time relationship in $SI and $FN fields is as follows:

$$U << t^E_{op}$$
$$t^W_{src} << t^E_{op} \quad (14)$$

where the relationship between U and t^W_{src} is not defined clearly, because these are dependent upon their initial states.

Timestamp Analysis : This operation replace an existing file with a file with the same name. The meaning of "Source File Left" is to leave a source file unchanged, and the existing target file is overwritten with the source file with the same name. After the operation completed, two field of $SI are changed, i.e., "Write time" is changed to t^W_{src}, and "MFT entry modified time" to t^E_{op}. Other fields remain unchanged. The meaning of "unchanged" is that timestamps of the file to be overwritten located in the target position remain in a new overwritten file. The order of U and t^W_{src} is not clear as in (14). It is difficult to decide the timestamps pattern is the result of "Overwrite with source file left" operation with only current timestamps, but if additional information provided, i.e. past timestamps of the source and the target file, we can decide the "Overwrite with source file left" operation is executed clearly, because there is no other file operations with same timestamp pattern in listed operations.

3.1.8 Overwrite with Source File Deleted

When "Overwrite with Source File Deleted" is performed, "Creation time" is changed to t^C_{src}, "Write time" to t^W_{src}, and "MFT entry modified time" to t^E_{op}, but "Access time" remains unchanged in $SI. And all of $FN fields remains unchanged.

$$\$SI(t^C_{src}, t^W_{src}, t^E_{op}, U^A)$$
$$\$FN(U^C, U^W, U^E, U^A) \quad (15)$$

The time relationship in $SI and $FN fields is as follows:

$$U << t^E_{op}$$
$$t^C_{src} << t^E_{op}$$
$$t^W_{src} << t^E_{op} \quad (16)$$

where the relationship between U, t^C_{src} and t^W_{src} is not defined clearly, because these are dependent upon their initial states.

Timestamp Analysis : This operation replace an existing file with a file with the same name, but a source file is to be deleted after the operation completed. Three field of $SI are changed, i.e. "Create time" is changed to t^C_{src}, "Write time" to t^W_{src} and "MFT entry modified time" to t^E_{op}. Other fields include "Access time" in $SI and all of four timestamp in $FN attribute remain unchanged. The order of U, t^C_{src} and t^W_{src} is not clear as in (16). It is difficult to decide what type of a file operation is executed with only current timestamps, but if additional information provided, i.e. past timestamps of the source and the target file, we can decide the "Overwrite with source file deleted" operation is executed clearly, because

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there is no other same timestamp pattern as the pattern of "Overwrite with source file deleted" operation.

3.1.9 File Name Change

When "File Name Change" is performed, "MFT entry modified time" field is changed to t^E_{op} in $SI with the remaining three timestamps are unchanged. Four timestamps in $FI are changed to timestamps in $SI of a source file. In this case, a source file and a destination file are same, i.e., $U^C=t^C_{SIsrc}$, $U^W=t^W_{SIsrc}$, and $U^A=t^A_{SIsrc}$, but $t^E_{op}\neq t^E_{SIsrc}$, respectively.

$$\$SI(U^C, U^W, t^E_{op}, U^A)$$
$$\$FN(t^C_{SIsrc},t^W_{SIsrc},t^E_{SIsrc},t^A_{SIsrc}) \quad (17)$$

The time relationship in $SI and $FN fields are as follows:

$$(U = t_{SIsrc}) << t^E_{op} \quad (18)$$

Timestamp Analysis : This operation changes a file name after the operation completed. In this case, there is no change of the file lcoation, so a source and a destination of file are the same file. In $FN attribute of (17), $t^C_{SIsrc},t^W_{SIsrc},t^E_{SIsrc}$ and t^A_{SIsrc} copied from its $SI attribute before the operation is performed. As in (18), ($U = t_{SIsrc}$) is a distinctive pattern, but it has similar pattern of "Move with a volume" operation in (9), unfortunately, this pattern is not unique.

3.1.10 File Attribute Change

When "File Attributes Change" is performed, "MFT entry modified time" field in $SI is changed to t^E_{op}. But the rest of fields are unchanged.

$$\$SI(U^C, U^W, t^E_{op}, U^A)$$
$$\$FN(U^C, U^W, U^E, U^A) \quad (19)$$

The time relationship in $SI and $FN fields is as follows:

$$U << t^E_{op} \quad (20)$$

Timestamp Analysis : This operation changes a file attribute, include Archive, Read-only, Hidden and file indexing attribute, after the operation completed. As in (19), this is a distinctive pattern, but it is difficult to decide what type of a file operation is executed with only current timestamps, because it is not known initial status of each timestamp field. But if additional information provided, i.e., past timestamps of the source file, we can decide the "File Attributes Change" operation is executed distinctly, because there is no other same timestamp pattern as of the operation.

3.2. Tables

We analyzed ten patterns of file operations in Sec. 3.1. In this section, we marked three types in a table of ten operations. The operation with a circle mark "○" means it can be identified by its distinct timestamp feature intuitively, the operations with a triangle mark "△" mean that it needed past timestamp to identify the file operation clearly, and operation with a cross mark "×" means that it has ambiguity with similar timestamp patterns.

Table 1. Intuition from timestamp patterns

Operations	$SI/$FN attributes patterns	Forensics by Intuition
Creation	$SI($t^C_{op}$, t^W_{op}, t^E_{op}, t^A_{op}) $FN($t^C_{op}$, t^W_{op}, t^E_{op}, t^A_{op})	○
Copy	$SI($t^C_{op}$, t^W_{src}, ($t^E_{op}+\Delta$), t^A_{op}) $FN($t^C_{op}$, t^W_{op}, t^E_{op}, t^A_{op})	○
Update-general file	$SI($U^C$, ($t^W_{op}+\Delta$), t^E_{op}, U^A) $FN($U^C$, U^W, U^E, U^A)	○
Update-MS Word/Office	$SI($U^C$, t^W_{op}, t^E_{op}, t^A_{op}) $FN($U^C$, t^W_{op}, t^E_{op}, t^A_{op})	○
Move within a Volume	$SI($U^C$, U^W, t^E_{op}, U^A) $FN($t^C_{src}$, t^W_{src}, t^E_{src}, t^A_{src})	×
Move to an Outside Volume	$SI($U^C$, U^W, ($t^E_{op}+\Delta$), t^A_{op}) $FN($t^C_{op}$, t^W_{op}, t^E_{op}, t^A_{op})	○
Overwrite with Source File Left	$SI($U^C$, t^W_{src}, t^E_{op}, U^A) $FN($U^C$, U^W, U^E, U^A)	△
Overwrite with Source File Deleted	$SI($t^C_{src}$, t^W_{src}, t^E_{op}, U^A) $FN($U^C$, U^W, U^E, U^A)	△
File Name Change	$SI($U^C$, U^W, t^E_{op}, U^A) $FN($t^C_{SIsrc}$,$t^W_{SIsrc}$,$t^E_{SIsrc}$,$t^A_{SIsrc}$)	×
File Attribute Change	$SI($U^C$, U^W, t^E_{op}, U^A) $FN($U^C$, U^W, U^E, U^A)	△

4. Case Analysis

4.1. Case I : Copy

In this case, we consider the copy operation case. A source file named "AfterPass.hwp" copied to a new name "AfterPassCopy.hwp". Fig. 1 shows $MFT header of the copied file. Only t^W_{src}, in $SI has a different timestamp, i.e. 2014-02-13 03:24:53. We can decide the case as a "Copy" operation applied to a source file intuitively, becase only t^W_{src}, in $SI has a precede timestamp and other timestamps are all to the same time as 2014-04-02 16:01:32. This pattern is unique in ten operations. The timestamps are as the followings:

Copied File : "AfterPassCopy.hwp"
-$Standard_Information
C: 2014-04-02 16:01:32 W:2014-02-13 03:24:53
E: 2014-04-02 16:01:32 A: 2014-04-02 16:01:32
-$File_Name
C: 2014-04-02 16:01:32 W: 2014-04-02 16:01:32

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E: 2014-04-02 16:01:32 A: 2014-04-02 16:01:32

Source File : “AfterPass.hwp”
-$Standard_Information
C: 2014-02-11 05:11:42 W: 2014-02-13 03:24:53
E: 2014-02-13 03:25:05 A:2014-02-11 05:11:42
-$File_Name
C: 2014-02-11 05:11:42 W:2014-02-11 05:11:42
E: 2014-02-11 05:11:42 A:2014-02-11 05:11:42

```
46 49 4C 45 30 00 03 00  44 37 62 15 02 00 00 00  FILE0...D7b.....
98 00 02 00 38 00 01 00  E0 01 00 00 00 04 00 00  ˜...8...à.......
00 00 00 00 00 00 00 00  04 00 00 00 DF 89 00 00  ............ß‰..
02 00 00 00 00 00 00 00  10 00 00 00 60 00 00 00  ............`...
00 00 00 00 00 00 00 00  48 00 00 00 18 00 00 00  ........H.......
4F 03 BE D0 8C 4E CF 01  D0 EA 48 29 6B 28 CF 01  O.¾Ð‰NÏ.ÐêH)k(Ï.
ED D3 C1 D0 8C 4E CF 01  4F 03 BE D0 8C 4E CF 01  íÓÁÐ‰NÏ.O.¾Ð‰NÏ.
20 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ...............
00 00 00 00 B2 02 00 00  00 00 00 00 00 00 00 00  ....²...........
38 1E 29 7F 00 00 00 00  30 00 00 00 78 00 00 00  8.)•....0...x...
00 00 00 00 00 00 03 00  5A 00 00 00 18 00 01 00  ........Z.......
0A 3F 00 00 00 00 02 00  4F 03 BE D0 8C 4E CF 01  .?......O.¾Ð‰NÏ.
4F 03 BE D0 8C 4E CF 01  4F 03 BE D0 8C 4E CF 01  O.¾Ð‰NÏ.O.¾Ð‰NÏ.
4F 03 BE D0 8C 4E CF 01  00 C0 00 00 00 00 00 00  O.¾Ð‰NÏ..À......
00 00 00 00 00 00 00 00  20 00 00 00 00 00 00 00  ........ .......
0C 02 41 00 46 00 54 00  45 00 52 00 50 00 7E 00  ..A.F.T.E.R.P.~.
32 00 2E 00 48 00 57 00  50 00 79 00 2E 00 68 00  2...H.W.P.y...h.
```

Figure 2. Case 1: Copied File

4.2. Case II :Content Update of General File

In this case, we consider the content update of general file operation case. A source file named “sample.txt” updated. Fig. 2 shows $MFT header of the updated file. Only t^{W}_{src}, in $SI has a different timestamp, i.e. 2014-02-13 03:24:53. We can decide the case as a “Copy” operation applied to a source file intuitively, becase only t^{W}_{src}, in $SI has a precede timestamp and other timestamps are all to the same time as 2014-04-02 16:01:32. This pattern is unique in ten operations. The timestamps are as the followings:

```
46 49 4C 45 30 00 03 00  44 B7 15 16 02 00 00 00  FILE0...D·......
87 00 01 00 38 00 01 00  78 01 00 00 00 04 00 00  ‡...8...x.......
00 00 00 00 00 00 00 00  05 00 00 00 7E F5 00 00  ............~õ..
22 00 00 00 00 00 00 00  10 00 00 00 60 00 00 00  "...........`...
00 00 00 00 00 00 00 00  48 00 00 00 18 00 00 00  ........H.......
07 E9 EF A4 C5 24 CF 01  96 CD 9B 71 9C 4E CF 01  .éï¤Å$Ï.Í›q‰NÏ.
96 CD 9B 71 9C 4E CF 01  07 E9 EF A4 C5 24 CF 01  Í›q‰NÏ..éï¤Å$Ï.
20 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ...............
00 00 00 00 C7 05 00 00  00 00 00 00 00 00 00 00  ....Ç...........
B8 E7 39 7F 00 00 00 00  30 00 00 00 70 00 00 00  ¸ç9•....0...p...
00 00 00 00 00 00 03 00  56 00 00 00 18 00 01 00  ........V.......
05 00 00 00 00 00 05 00  07 E9 EF A4 C5 24 CF 01  .........éï¤Å$Ï.
07 E9 EF A4 C5 24 CF 01  07 E9 EF A4 C5 24 CF 01  .éï¤Å$Ï..éï¤Å$Ï.
07 E9 EF A4 C5 24 CF 01  10 00 00 00 00 00 00 00  .éï¤Å$Ï........
0E 00 00 00 00 00 00 00  20 00 00 00 00 00 00 00  ........ .......
0A 03 73 00 61 00 6D 00  70 00 6C 00 65 00 2E 00  ..s.a.m.p.l.e...
74 00 78 00 74 00 00 00  40 00 00 00 28 00 00 00  t.x.t...@...(...
```

Figure 2. Case 2: Updated File

Updated File : “sample.txt”
-$Standard_Information
C: 2014-02-08 12:02:31 W:2014-04-02 17:53:24
E: 2014-04-02 17:53:24 A: 2014-02-08 12:02:31
-$File_Name
C: 2014-02-08 12:02:31 W: 2014-02-08 12:02:31
E: 2014-02-08 12:02:31 A: 2014-02-08 12:02:31

Source File : “sample.txt”
C: 2014-02-08 12:02:31 W: 2014-02-08 12:02:31
E: 2014-02-08 17:53:24 A: 2014-02-08 12:02:31
-$File_Name
C: 2014-02-08 12:02:31 W: 2014-02-08 12:02:31
E: 2014-02-08 12:02:31 A: 2014-02-08 12:02:31

4.3. Case III: Move within a Volume & Finename Change

In this case, we compare “Move within a Volume” operation and “Filename Change” operation. Even if they have distinguishable timestamp patterns to other eight operations, the two operations make same timestamp patterns, so it is unfortunately difficult to distinguish between the two. The two operated files, i.e. “sample1.txt” and “sample2.txt” are copied from the same source file but the copy operation time was different; “sample1.txt” copied at 2014-04-06 04:50:49 and “sample2 .txt” copied at 2014-04-06 09:00:48.

The “sample1.txt” moved from “c:\sample1.txt” to “c:\User\cho\Desktop\sample1.txt”. The moved file’s $MFT entry header and two attributes are shown in Fig. 3.

The “sample2.txt” changed its filename to “sample2-fNameChg.txt”. The file’s $MFT entry header and two attributes are shown in Fig. 4.

In “sample1.txt”, the “Create time”, “Write time”, and “Access time” of $SI are same to “Create time”, “Write time”, and “Access time” of $FN, respectively. The “sample2.txt” have same timestamps pattern as “sample1.txt”. If the two operation can be distinguished intuitively, it is need an additional information about each operation.

```
46 49 4C 45 30 00 03 00  0E CA 78 1D 02 00 00 00  FILE0....Êx.....
28 00 01 00 38 00 01 00  50 01 00 00 00 04 00 00  (...8...P.......
00 00 00 00 00 00 00 00  04 00 00 00 EC 26 01 00  ............ì&..
03 00 00 00 00 00 00 00  10 00 00 00 60 00 00 00  ............`...
00 00 00 00 00 00 00 00  48 00 00 00 18 00 00 00  ........H.......
96 E0 E8 C7 53 51 CF 01  96 CD 9B 71 9C 4E CF 01  ‰àèÇSQÏ.Í›q‰NÏ.
6D 6B 7D F0 65 51 CF 01  96 E0 E8 C7 53 51 CF 01  mk}ðeQÏ.‰àèÇSQÏ.
20 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ...............
00 00 00 00 77 07 00 00  00 00 00 00 00 00 00 00  ....w...........
E0 74 E9 80 00 00 00 00  30 00 00 00 70 00 00 00  àté€....0...p...
00 00 00 00 00 00 03 00  58 00 00 00 18 00 01 00  ........X.......
0A 3F 00 00 00 00 02 00  96 E0 E8 C7 53 51 CF 01  .?......‰àèÇSQÏ.
96 CD 9B 71 9C 4E CF 01  F6 41 EB C7 53 51 CF 01  ‰Í›q‰NÏ.öAëÇSQÏ.
96 E0 E8 C7 53 51 CF 01  28 00 00 00 00 00 00 00  ‰àèÇSQÏ.(.......
27 00 00 00 00 00 00 00  20 00 00 00 00 00 00 00  '....... .......
0B 03 73 00 61 00 6D 00  70 00 6C 00 65 00 31 00  ..s.a.m.p.l.e.1.
2E 00 74 00 78 00 74 00  80 00 00 00 40 00 00 00  ..t.x.t.€...@...
```

Figure 3. Case 3-1: Move within a Volume

The timestamps of “sample1.txt” are as the followings:

Moved File : “c:\User\cho\Desktop\sample1.txt”
-$Standard_Information
C: 2014-04-06 04:50:49 W:2014-04-02 17:53:24
E: 2014-04-06 07:00:48 A: 2014-04-06 04:50:49
-$File_Name

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C: 2014-04-06 04:50:49 W: 2014-04-02 17:53:24
E: 2014-04-06 04:50:49 A: 2014-04-06 04:50:49

Source File : “sample1.txt”
-$Standard_Information
C: 2014-04-06 04:50:49 W:2014-04-02 17:53:24
E: 2014-04-06 04:50:49 A: 2014-04-06 04:50:49
-$File_Name
C: 2014-04-06 04:50:49 W: 2014-04-06 04:50:49
E: 2014-04-06 04:50:49 A: 2014-04-06 04:50:49

The timestamps of “sample2.txt” are as the followings:

FileName Changed File : “c:\sample2-fNameChg.txt”
-$Standard_Information
C: 2014-04-06 09:00:48 W:2014-04-02 17:53:24
E: 2014-04-06 15:48:38 A: 2014-04-06 09:00:48
-$File_Name
C: 2014-04-06 09:00:48 W: 2014-04-02 17:53:24
E: 2014-04-06 09:00:48 A: 2014-04-06 09:00:48

Source File : “sample2.txt”
-$Standard_Information
C: 2014-04-06 09:00:48 W:2014-04-02 17:53:24
E: 2014-04-06 09:00:48 A: 2014-04-06 09:00:48
-$File_Name
C: 2014-04-06 09:00:48 W: 2014-04-06 09:00:48
E: 2014-04-06 09:00:48 A: 2014-04-06 09:00:48

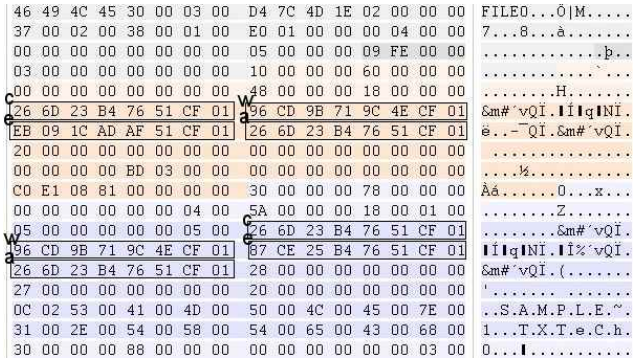

Figure 4. Case 3-2: Filename Change

5. Conclusion

In this paper, an intuitive computer forensic method by timestamp changing patterns of operations on file in Windows NTFS file system is proposed. Ten distinguishable patterns by their timestamp changes make a decision on identifying what kind of file operation is performed. And analyses on ten timestamp patterns are conducted for applying an intuitive timestamp forensic method only with current timestamps. We understood that some patterns are easily identified by their distinct timestamp feature intuitively, and some patterns are needed past timestamp to identify the file operation clearly, and some patterns have ambiguity with similar timestamp patterns.

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References

[1] Wikipedia, MAC times, http://en.wikipedia.org/wiki/MAC_times

[2] E. Casey, “Uncertainty and Loss in Digital Evidence”, International Journal of Digital Evidence, vol. 1:2, Summer 2002.

[3] Jonathan Grier, “Detecting data theft using stochastic forensics”, digital investigation, vol. 8, pp. S71-S77, 2011.

[4] C. Boyd and P. Forster, “Time and Date Issues in Forensic Computing − A Case Study”, Digital Investigation, vol. 1, no. 1, pp. 18-23, Feb. 2004.

[5] K.P. Chow et. al., “The Rules of Time on NTFS File System”, SADFE ‘07, pp. 71-85, March 2007.

[6] M. W. Stevens, “Unification of Relative Time Frames for Digital Forensics”, Digital Investigation, pp. 225-239, Jan. 2004.

[7] Willasen, S., “Hypothesis-based Investigation of Digital Timestamps”, IFIP International Federation for Information Processing, Vol. 285; Advances in Digital Forensics IV; pp. 75-86, 2008.

[8] Olsson J and Boldt M, “Computer forensic timeline visualization tool”, Digital Investigation, vol. 6, pp. S78−87. 2009.

[9] C. Hargreaves and J. Patterson, “An automated timeline reconstruction approach for digital forensic investigations”, Digital Investigation, vol. 9, S69−S79, 2012.

[10] Gyu-Sang Cho, “A computer forensic method for detecting timestamp forgery in NTFS“, computer & security, vol. 34, pp. 36-46, 2013.

[11] Kevin D. Fairbanks, “An analysis of Ext4 for digital forensics”, Digital Investigation, No. 9, pp. S118−S130, 2012.

[12] Ming Xu, et. al., “A metadata-based method for recovering files and file traces from YAFFS2”, Digital Investigation, no. 10 , pp. 62−72, 2013.

[13] A. Marrington, et. al., “CAT Detect (Computer Activity Timeline Detection): A tool for detecting inconsistency in computer activity timelines”, , digital investigation, no 8, pp. S52-S61, 2011.

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